5 Life: History and Origin

5.1 Properties of life

1. Organization: Being structurally composed of one or more cells, which are the basic units of life.
   - prokaryote: no nucleus
   - eukaryote: membrane bound nucleus.

2. Sensitivity: respond to stimuli.

3. Energy Processing

4. Growth and Development

5. Reproduction

   - hereditary mechanisms to make more of self; DNA based.
6. Regulation, including homeostasis.

7. Evolution.

5.2 Origin of life: 3 hypotheses

- Extraterrestrial origin (panspermia): meteor, comet borne from elsewhere in universe
  - evidence of amino acids and other organic material in space (but often both D & L forms)
  - questionable bacterial fossils in Martian rock

-However, this would imply that some other origin of life was likely because it would have had to happen elsewhere before it could be transported here, and the only difference would be that life did not originate on Earth.

- Spontaneous origin on earth: primitive self-replicating macromolecules acted upon by natural selection ((macro)Evolution is one example of this)

-This is often attacked for the seeming impossibility for life to have been produced by a chemical reaction triggered by lightning and the ability of any produced DNA to actually be in a sequence that could produce a working model of life if replicated. It is also attacked for religious reasons, as it bypasses things like the idea of a supreme being directly creating humans. It also seems unlikely to some that such huge changes are possible in evolution without evidence of an "in-between stage" that is credible. Many of the stages of man are disputed due to their somewhat shakey grounds. For example, bones from other animals have been taken accidentally in some cases to be part of a humanoid, and complete skeletons have been sketched out from a limited number of bones.

- Special creation: religious explanations (Intelligent Design is one popular example of this.) These explanations contend that life was created by God (or perhaps some other Intelligent Designer).
  - Proponents of Intelligent design suggest that the vast complexity of life could only have been intentionally designed while other creationists cite biblical support.

-This is often attacked for many of the same reasons that religion is attacked, and is often regarded as superstitious and/or unscientific.

- It is debated as to whether schools should teach one hypothesis or the other when talking about the origin of life. However, since they are all currently known major hypotheses (and sometimes hypotheses proven wrong are shown for educational purposes), this wikibook includes what it can without discriminating unfairly against one hypothesis or the other.

5.3 The early earth

It is believed that the Earth was formed about 4.5 billion years ago.

- Heavy bombardment by rubble ceased about 3.8 billion years ago.
- Reducing atmosphere: much free H
  - also H_2O, NH_3, CH_4
  - little, if any, free O_2
  - with numerous H electrons, require little energy to form organic compounds with C
- Warm oceans, estimated at 49-88°C
- Lack of O_2 and consequent ozone (O_3) meant considerable UV energy

Chemical reactions on early earth

- UV and other energy sources would promote chemical reactions and formation of organic molecules
- Testable hypothesis: Miller-Urey experiment
  - simulated early atmospheric conditions
  - found amino acids, sugars, etc., building blocks of life
  - won Nobel prize for work
  - experiment showed prebiotic synthesis of biological molecules was possible

Issues

- Miller later conceded that the conditions in his experiments were not representative of what is currently thought to be those of early earth
- He also conceded that science has no answer for how amino acids could self-organize into replicating molecules and cells
- In the 50 years since Miller-Urey, significant issues and problems for biogenesis have been identified. This is a weak hypothesis at this time.

- Conclusion: Life exists, we don't know why.

5.4 Origin of cells

Cells are very small and decompose quickly after death. As such, fossils of the earliest cells do not exist. Scientists have had to form a variety of theories on how cells (and hence life) was created on Earth.

- Bubble hypothesis
  - A. Oparin, J.B.S. Haldane, 1930's
- Primary abiogenesis: life as consequence of geochemical processes
- Protobionts: isolated collections of organic material enclosed in hydrophobic bubbles
  - Numerous variants: microspheres, protocells, protobionts, micelles, liposomes, coacervates
- Other surfaces for evolution of life
  - deep sea thermal vents
  - ice crystals
  - clay surfaces
  - tidal pools

5.5 The RNA world?

- DNA → RNA → polypeptide (protein)
- Catalytic RNA: ribozyme
  - discovered independently by Tom Cech and Sid Altman (Nobel prize)
  - catalytic properties: hydrolysis, polymerization, peptide bond formation, etc.
- Self-replicating RNA molecule may have given rise to life
  - consistent with numerous roles for RNA in cells as well as roles for ribonucleotides (ATP)
  - relationship to bubble-like structures is uncertain

5.6 The earliest cells

- Microfossils
  - ˜3.5 by
  - resemble bacteria: prokaryotes
  - biochemical residues
  - stromatolites
- Archaebacteria (more properly Archaea)
  - extremophiles: salt, acid, alkali, heat, methanogens
  - may not represent most ancient life
- Eubacteria
  - cyanobacteria: photosynthesis
- atmospheric O_2; limestone deposits

- chloroplasts of eukaryotes

Cyanobacteria

5.7 Major steps in evolution of life

- Prebiotic synthesis of macromolecules
- Self replication
  - RNA? (primitive metabolism)
- DNA as hereditary material
- 1st cells
- Photosynthesis
- Aerobic respiration
- Multicellularity (more than once)

This text is based on notes very generously donated by Dr. Paul Doerder, Ph.D., of the Cleveland State University.